VALLALA SAIKUMAR

Design Verification Engineer (Fresher) | Verilog | SystemVerilog | UVM | Functional Verification

Jangoan, Telangana, India | 9390140233 | vallalasaikumar6@gmail.com | linkedin.com/in/vallalasaikumar |
github.com/saikumarvallala6-crypto

PROFESSIONAL SUMMARY

Electronics and Communication Engineering graduate (2026) specializing in VLSI Design Verification. Hands-on experience building SystemVerilog testbenches and verifying RTL designs (I2C, UART) using constrained-random verification, functional coverage, and assertions. Proficient in Verilog, SystemVerilog, UVM, and simulation/debug using QuestaSim and Vivado. Seeking a Design Verification / RTL Engineer role to contribute to reliable, high-quality semiconductor solutions.

TECHNICAL SKILLS

Languages: Verilog HDL, SystemVerilog, C

Verification: UVM, Testbench Development (SV,UVM), Functional Verification, Constrained-Random Verification (CRV), Functional Coverage, Code Coverage, SystemVerilog Assertions (SVA)

Design: Digital Electronics, RTL Design, FSM Design

Protocols: I2C, UART

EDA Tools: QuestaSim, Xilinx Vivado, Xilinx ISE

Scripting / Tools / OS: TCL, Git, Windows.

PROJECTS

UART Design and Functional Verification using SystemVerilog

- Designed a UART controller in Verilog supporting configurable baud rate generation, framing (start/stop bits), and parity-based error detection.
- Built a layered SystemVerilog testbench (generator, driver, monitor, scoreboard) to verify transmitter and receiver functionality end-to-end.
- Applied constrained-random stimulus across data widths, baud rates, and parity configurations; wrote SVA to check frame timing and integrity.
- Developed functional coverage models and used coverage reports to close verification gaps; debugged via waveform analysis in QuestaSim under error-injected conditions.

Tech: Verilog, SystemVerilog, Functional Coverage, SVA, Testbench Development, QuestaSim, UART Protocol.

Design and Functional Verification of I2C Master Controller Using Verilog

- Designed and implemented an I2C Master Controller in Verilog supporting START/STOP conditions, slave addressing, read/write operations, and ACK/NACK handling.
- Developed RTL modules and testbenches to verify protocol functionality; performed simulation, debugging, and waveform analysis to validate data transfer and error conditions.

Tech: Verilog HDL, Functional Verification, Testbench Development, Digital Design, I2C Protocol.

WORK EXPERIENCE

Design Verification Trainee | *VLSI FIRST Institute, Hyderabad, Telangana, India* **Dec 2025 - Present**

- Designed and executed SystemVerilog testbenches to verify RTL designs, achieving 90%+ functional coverage through constrained-random verification scenarios.
- Performed simulation, debugging, and waveform analysis using QuestaSim to identify and resolve design bugs at the module level; wrote and maintained verification test plans and coverage matrices.

EDUCATION

Bachelor of Technology, Electronics and Communication Engineering **Oct 2022 - May 2026**

Vaagdevi Engineering College, Warangal, Telangana | CGPA: 8.32

CERTIFICATIONS & ACHIEVEMENTS

- Front End Design and Its Applications using Xilinx - Industry-Oriented Mini Project, 8-week internship.
- 1st Place, Electronics Technical Quiz organized by the IETE Student Chapter.